

Computer Networks

Lecture 7: Switching

Content



- **Circuit Switching**
- **Packet Switching**

Virtual Circuit Network: 3-Phases

A virtual-circuit network has three phases: setup, data transfer, and teardown, similar to a circuit-switched network.

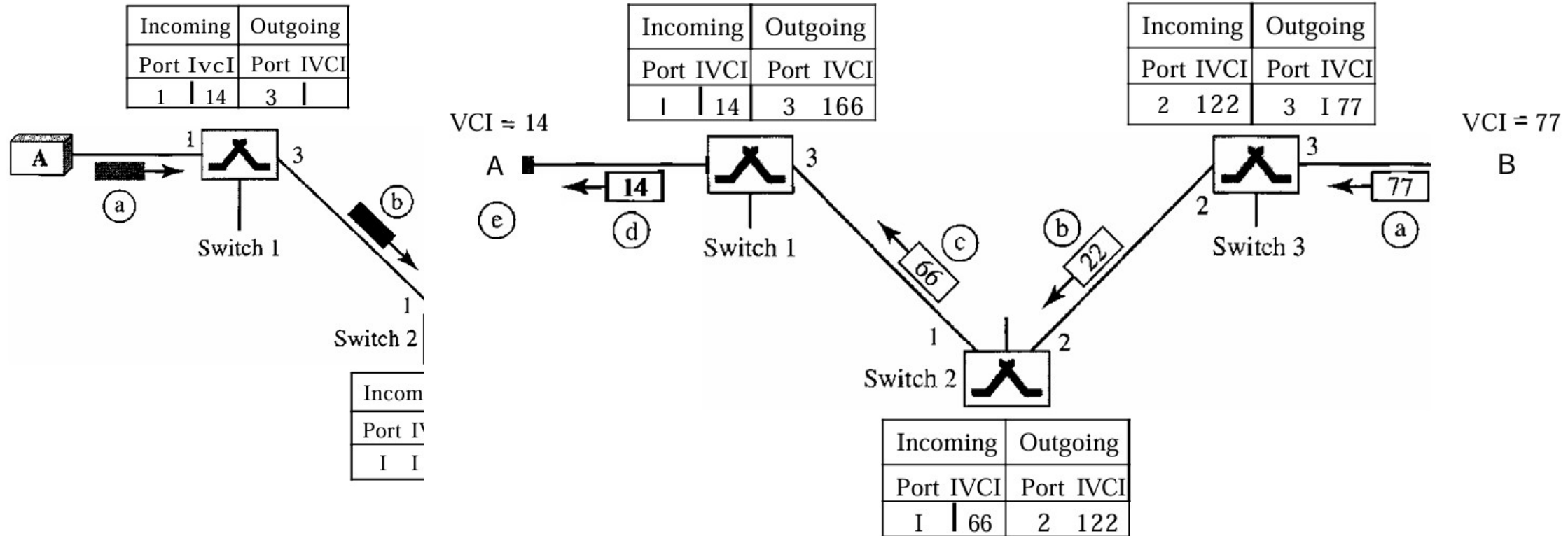
- During the setup phase, the source and destination use their global addresses to create entries in the switches' tables.
- Data transfer takes place after the connection is established.
- In the teardown phase, the source and destination notify the switches to remove the connection entries from their tables.

VCN: Setup Phase

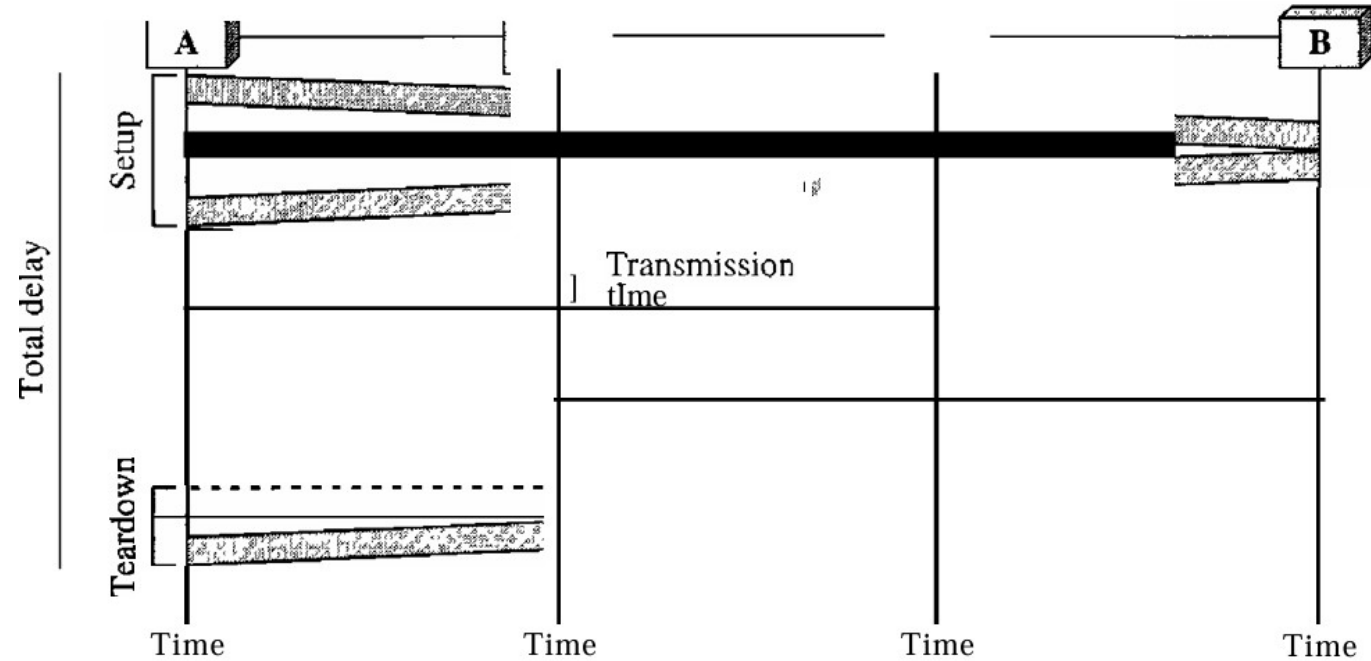
Steps of Setup phase is as follows

- The source first sends a setup request to the destination using its global address.
- Each intermediate switch receives this request and creates a table entry for the connection.
- The switch assigns a local identifier (VCI) for the incoming and outgoing links.
- These table entries together establish a fixed path from source to destination.
- Once the destination accepts the request, a confirmation is sent back, and the connection is ready for data transfer.

VCN Setup: Request



VCN : Delay



$$\text{Total delay} = 3T + 3\tau + \text{setup delay} + \text{teardown delay}$$

Question

1. A circuit-switched network has 20 channels, each with a data rate of 64 kbps.
If a call occupies one channel for 5 minutes, calculate:
 - A. Total data carried by one call
 - B. Maximum number of simultaneous call

2. A circuit-switched exchange has 30 circuits. Each call requires 1 circuit and lasts 3 minutes. If 15 calls arrive per minute, find:
 - A. Whether the system is blocking or non-blocking
 - B. Maximum number of calls served in 10 minutes

3. If a packet-switched network has a packet loss probability of 0.01, find the probability that all packets arrive successfully when 10 packets are sent.

Switch Structure

Circuit switching today can use either of two technologies:

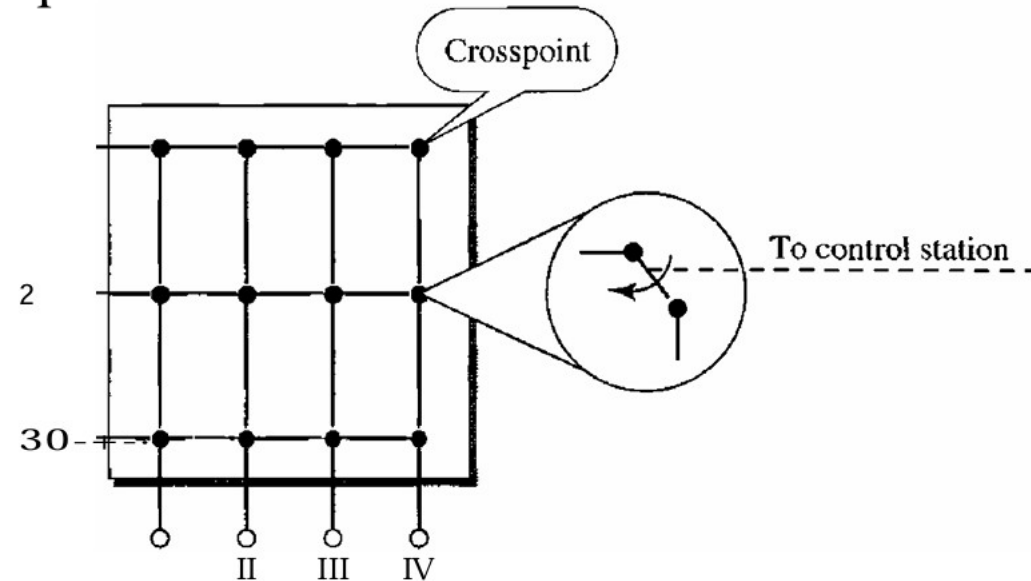
- the space-division switch or
- the time-division switch

In space-division switching, the paths in the circuit are separated from one another spatially.

- Crossbar Switch
- Multistage Switch

Crossbar Switch

A crossbar switch connects n inputs to m outputs in a grid, using electronic microswitches (transistors) at each crosspoint



The major limitation of this design is the number of crosspoints required. To connect n inputs to m outputs using a crossbar switch requires $n \times m$ crosspoints.

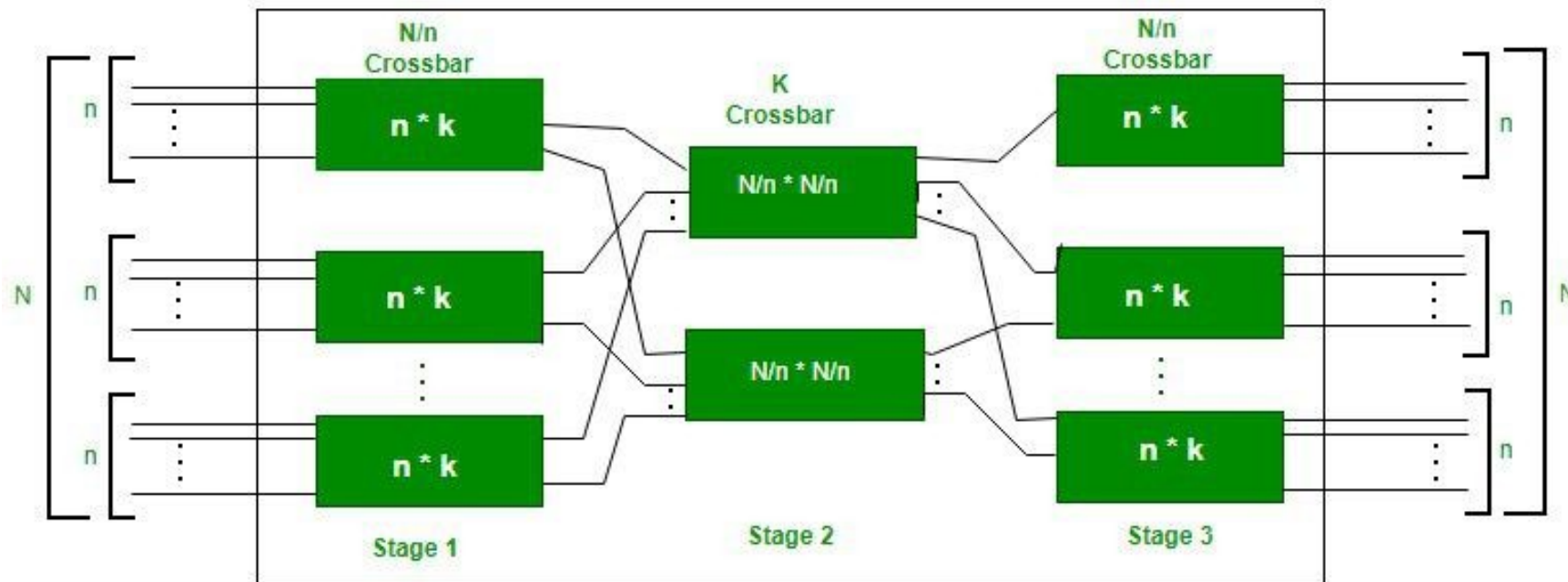
Multistage Switch: Design

In a single crossbar switch, only one row or column (one path) is active for any connection. So we need $N \times N$ crosspoints. If we can allow multiple paths inside the switch, we can decrease the number of crosspoints.

1. We divide the N input lines into groups, each of n lines. For each group, we use one crossbar of size $n \times k$, where k is the number of crossbars in the middle stage. In other words, the first stage has N/n crossbars of $n \times k$ crosspoints.
2. We use k crossbars, each of size $(N/n) \times (N/n)$ in the middle stage.
3. We use N/n crossbars, each of size $k \times n$ at the third stage.

Multistage Switch

In a single crossbar switch, only one row or column (one path) is active for any connection. So we need $N \times N$ crosspoints. If we can allow multiple paths inside the switch, we can decrease the number of crosspoints.



Question

A three-stage space-division switching network is to be designed for 200 input lines and 200 output lines. The network uses a multistage architecture with 4 middle-stage switches, and each input-stage switch handles 20 input lines.

- Determine the number of switches in each stage of the network.
- Find the total number of switches required in the three-stage switch.
- Calculate the total number of crosspoints required in the three-stage switching network and compare it with a single-stage 200×200 space-division switch by finding the percentage reduction in crosspoints.

Multistage Switch

Total no. of crosspoint

Stage 1:

There are (N/n) crossbars, each of size $(n \times k)$. Total: $(\frac{N}{n}) \times (n \times k) = kN$

Stage 2:

There are (k) crossbars, each of size $(\frac{N}{n}) \times (\frac{N}{n})$. Total: $(k \times (\frac{N}{n})^2)$

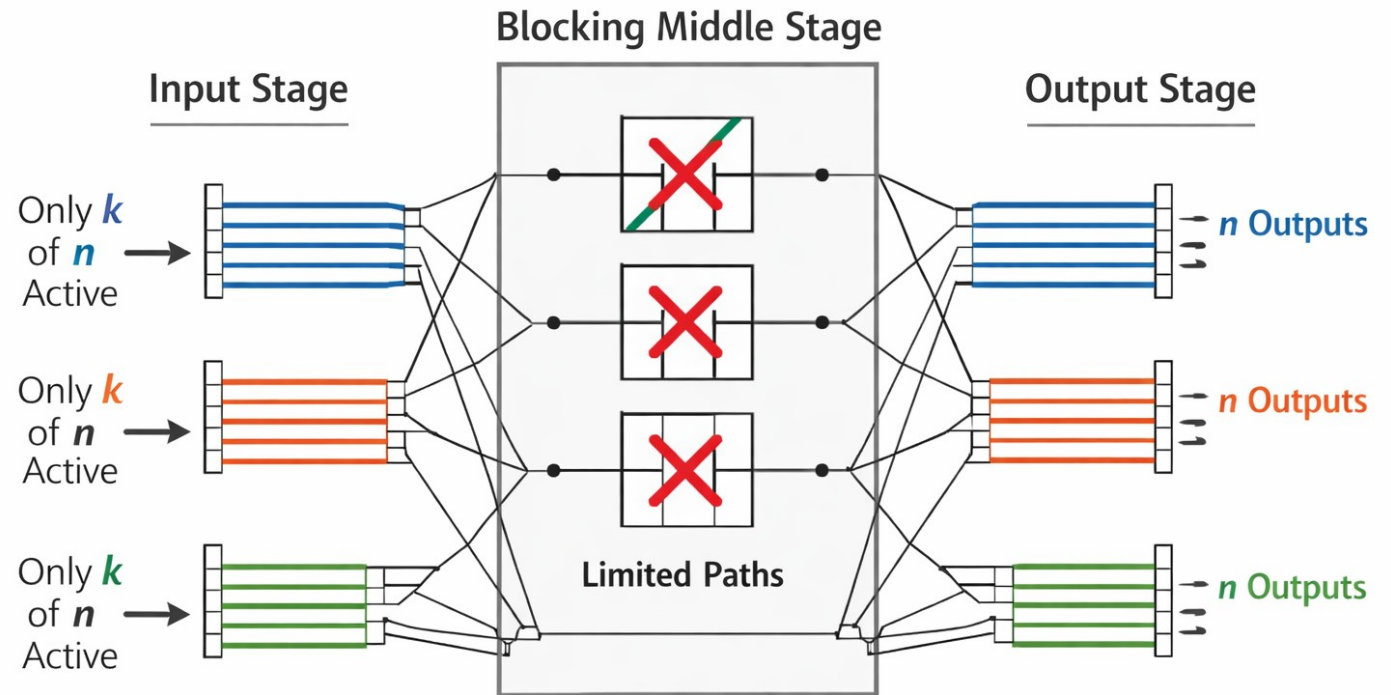
Stage 3:

There are (N/n) crossbars, each of size $(k \times n)$. Total: $(\frac{N}{n}) \times (k \times n) = kN$

Total: $2kN + k \left(\frac{N}{n}\right)^2$

Multistage Switch: Blocking

In the multistage switch only k of the first n inputs can use the switch at a time, only k of the second n inputs can use the switch at a time, and so on. The small number of crossbars at the middle stage creates blocking.



*Only k out of n can connect from each group, causing **Blocking**.*

Multistage Switch: Blocking

Clos investigated the condition of nonblocking in multistage switches and came up with the following formula. In a nonblocking switch, the number of middle-stage switches must be at least $2n - 1$

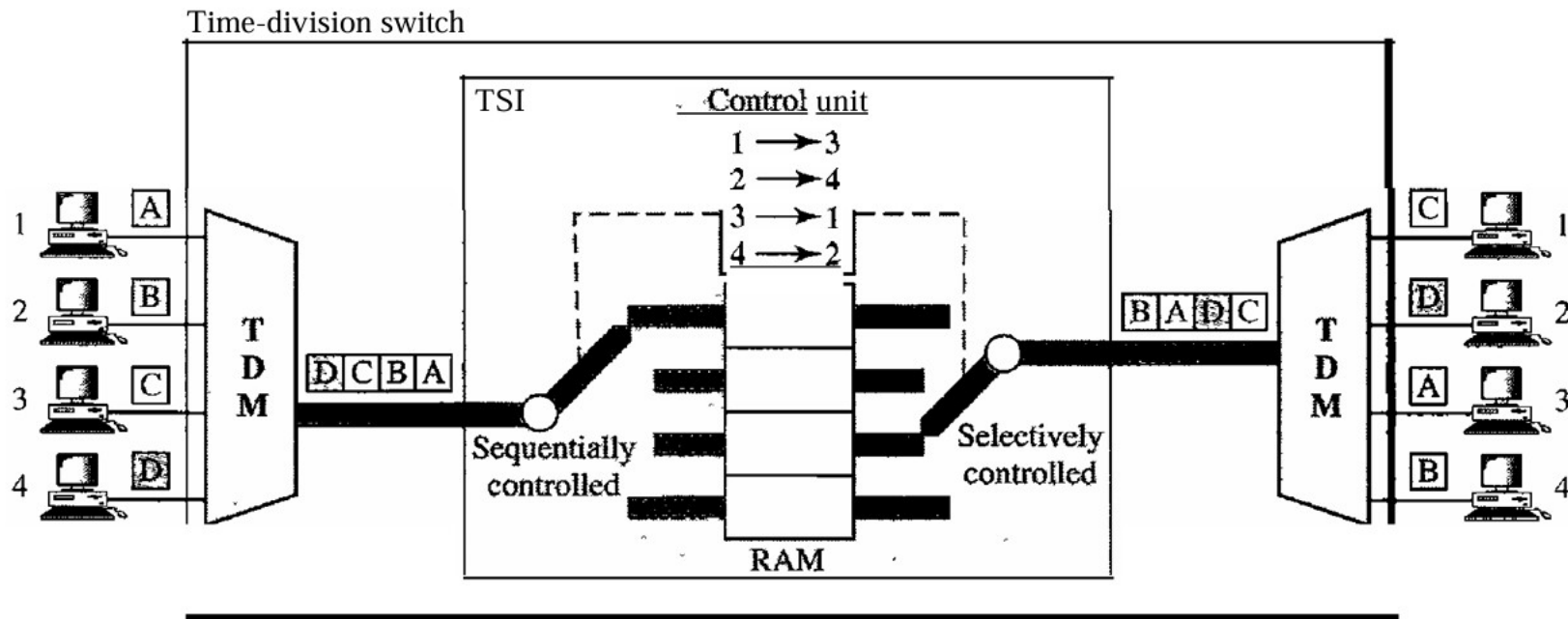
$$C = 2Nk + \frac{kN^2}{n^2}$$

Substitute $k = 2n - 1$

$$C = 2N(2n - 1) + \frac{(2n - 1)N^2}{n^2}$$

Time Division Switch

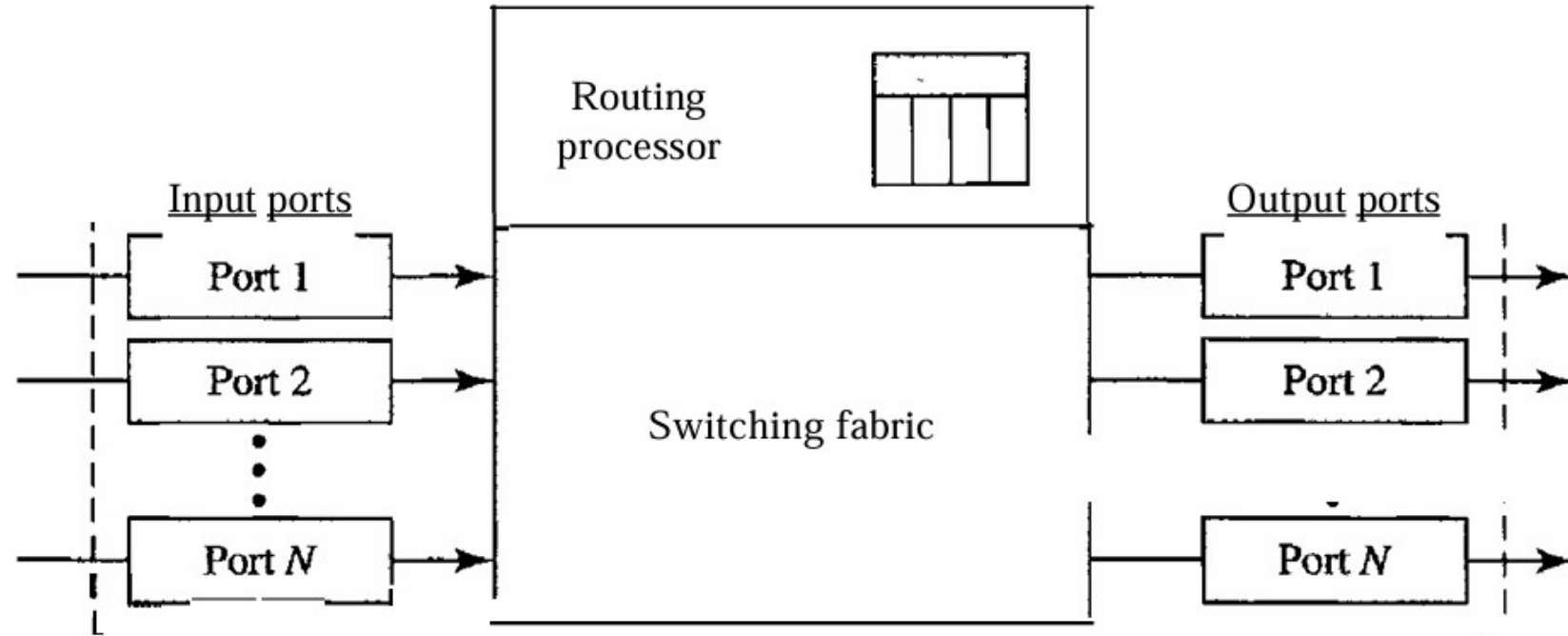
Time-division switching uses time-division multiplexing (TDM) inside a switch.



Packet Switches

A switch used in a packet-switched network has four components:

- input ports,
- output ports,
- the routing processor, and
- the switching fabric



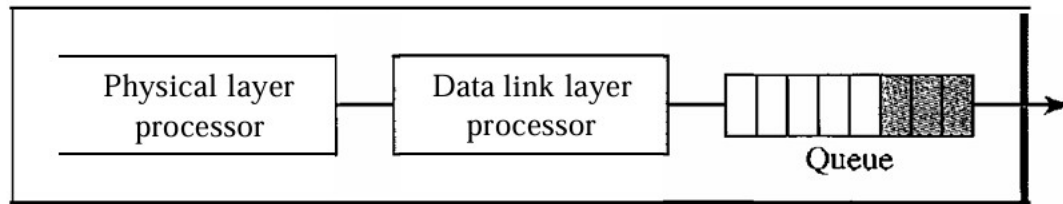
Packet Switches: Ports

Input Port

An input port performs the physical and data link functions of the packet switch.

- The bits are constructed from the received signal.
- The packet is decapsulated from the frame.
- Errors are detected and corrected.
- The packet is now ready to be routed by the network layer.

Input port

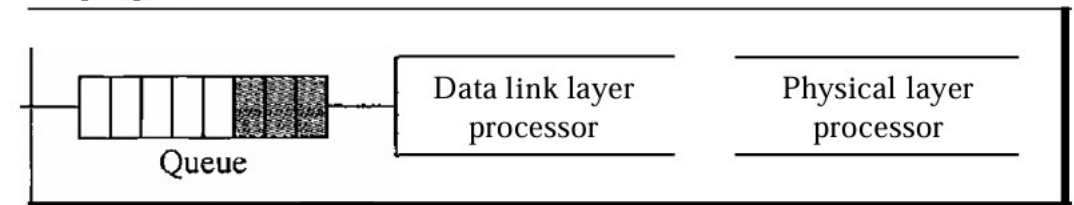


Output Port

output port performs the same functions as the input port, but in the reverse order

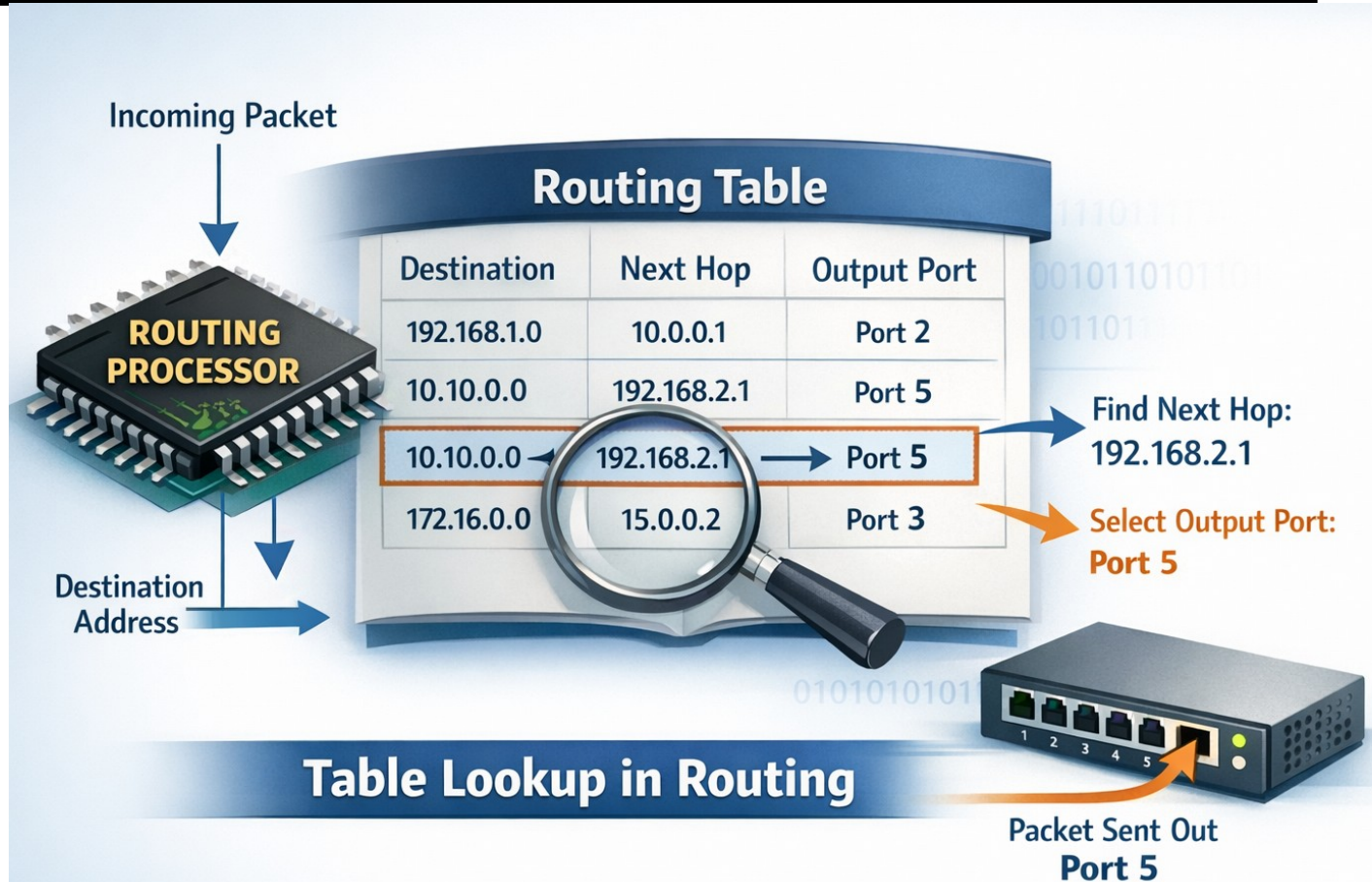
- Outgoing packets are queued,
- The packet is encapsulated in a frame, and
- Finally, the physical layer functions are applied to the frame to create the signal to be sent on the line.

Output port



Packet Switches: Routing Processor

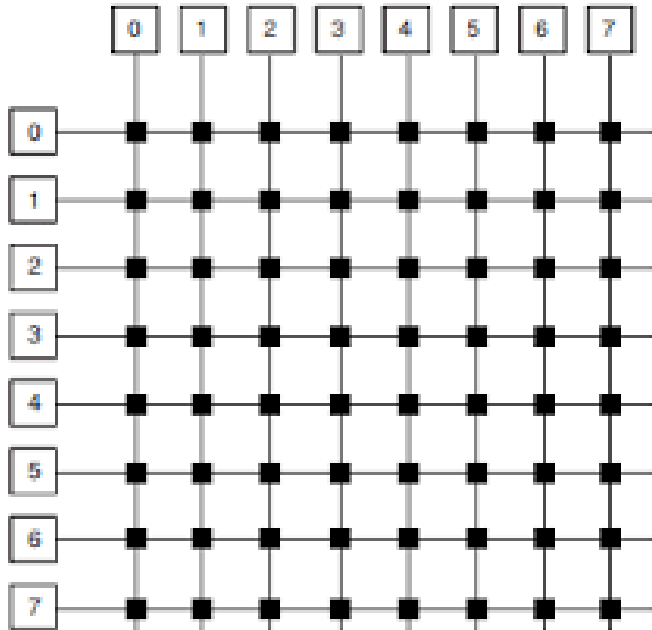
The routing processor works at the network layer. It uses the destination address to look up the routing table, find the next-hop address, and select the appropriate output port for forwarding the packet.



Packet Switches: Switching Fabrics

Packet switches are specialized mechanisms that use a variety of switching fabrics.

Crossbar Switch: The simplest type of switching fabric is the crossbar switch



Banyan Switch: A banyan switch is a multistage switch with microswitches at each stage.

